COUNTRY PROFILES

Nuclear Power in Poland

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- Poland plans to have nuclear power from about 2036 as part of a diverse energy portfolio, moving it away from heavy dependence on coal.
- Poland earlier considered a stake in the planned Visaginas nuclear power plant in Lithuania.
- Polls show strong support among the public for the construction of the country's first nuclear power plant.

0	0	0
OPERABLE REACTORS	REACTORS UNDER CONSTRUCTION	REACTORS SHUTDOWN
0 MWe	0 MWe	0 MWe

Electricity sector

Total generation (in 2024): 169 TWh

Generation mix: coal 94.6 TWh (56%); wind 24.8 TWh (15%); natural gas 21.0 TWh (12%); solar 15.2 TWh (9%); biofuels & waste 8.2 TWh (5%); hydro 3.1 TWh (2%); oil 2.3 TWh.

Privacy - Terms

Import/export balance: 2.0 TWh net import (15.2 TWh imports; 13.2 TWh exports)

Total consumption: 134 TWh

Per capita consumption: c. 3700 kWh in 2024

Source: International Energy Agency and The World Bank. Data for year 2024.

Per capita electricity consumption in Poland has increased from about 3200 kWh in 2000 to about 3700 kWh in 2024.

Coal is the dominant source of power in Poland. In 1990 it provided over 95% of the country's electricity, but significant investment in wind and solar capacity since, and a shift to natural gas, has reduced that figure to 60%.

Energy policy

Poland's energy demand is growing significantly. Total final consumption has grown by over 25% since 1990.

Poland is the second largest coal producer in Europe, behind Germany. In 2022, 42% of the country's primary energy consumption was coal, and all fossil fuels together accounted for 87%1. Poland produces modest quantities of oil and natural gas, but is a significant net importer of both fuels, primarily from Russia.

Despite its plentiful domestic resources of coal, the European Union's (EU's) strict climate policy targets mean the country is diversifying away from coal. The Energy Policy of Poland until 2040 (EPP2040), adopted by the cabinet in February 2021, sets out the framework for this energy transition in Poland. Key elements include:

- Share of coal in electricity generation limited to 56% in 2030.
- Natural gas will be a ‘bridge fuel’.
- Share of final energy consumption from renewable sources will be at least 23% in 2030, and share of electricity from renewable sources will be at least 32% in 2030.
- The country’s first nuclear power reactor with a capacity of 1-1.6GW will be commissioned in 2033*. Further reactors will be commissioned every 2-3 years up to a total of six.

* An updated schedule was announced in December 2024, which envisages commissioning of the country's first reactor in 2036.

In 2009 shale gas exploration and the nuclear programme were the immediate priorities. However, ExxonMobil (in 2012) and Chevron (in 2015) pulled out of exploration of shale gas in Poland after initial results proved disappointing, and there has been little activity since. In 2011 the International Energy Agency showed Poland's shale gas resources as the largest in Europe, at over 5000 billion cubic metres and the government quoted reserves of over 750 billion m3 as it prepared to release new regulations to encourage exploration and extraction of those. More recent estimates have slashed the shale gas resource estimates to one-tenth of earlier figures. The United Stated Geological Survey (USGS) in 2012 estimated 38 billion m3 recoverable.

Regional transmission links

In March 2015 an agreement was signed by Ukraine's Ukrenergo distribution company and Polenergia, a Polish counterpart, to export electricity as part of the Ukraine-EU 'energy bridge', and related to the Baltic Energy Market Interconnection Plan. The plan was to enable greater use of Ukraine's nuclear capacity and generate funds to pay for increasing that capacity. A 750 kV transmission connection from Khmelnistki to Rzeszow in Poland that closed in the 1990s was to be reinstated, with Khmelnistki 2 then being disconnected from the Ukraine grid and synchronized with the EU grid. In 2020 it appeared that the plan would not proceed, but work recommenced in the wake of Russia's military offensive in Ukraine and was completed in 2023.

Apart from earlier plans for possible Polish participation in the Baltic states nuclear plant (see Lithuanian liaison section in Appendix), the high voltage (400 kV) 1000 MW LitPol Link project, completed in February 2025, helps integrate the power grid of the three Baltic states with the synchronous system of Europe and improves transmission capacity between Lithuania and Poland.

Nuclear power plans

Poland has plans for both large and small reactors.

Reactors planned in Poland

Location / site	Type	Gross capacity (MWe)	Construction start
Lubiatowo-Kopalino	AP1000	3x1250	2028
Włocławek	BWRX-300	4x300	2028

Subtotal planned: 3 large (3750 MWe) 4 small (1200 MWe)

Reactors proposed in Poland

Location / site	Type	Gross capacity (MWe)
Pątnów	APR1400	2x1400
Dąbrowa Górnicza	BWRX-300	4x300
Nowa Huta	BWRX-300	4x300
Ostrołęka	BWRX-300	4x300
Stawy Monowskie	BWRX-300	4x300
Tarnobrzeg Special Economic Zone	BWRX-300	4x300

Large reactors

The Polish cabinet decided early in 2005 that for energy diversification, and to reduce carbon dioxide and sulfur emissions, the country should move immediately to introduce nuclear power, so that an initial plant might be operating soon after 2020. A 2009 report to the Ministry of the Economy identified nuclear as the most cost-effective method of carbon dioxide abatement of the major generating options. A resolution by the council of ministers then called for the construction of at least two plants in Poland, or at least 4.6 GWe out of a predicted 52 GWe total capacity – to provide 15% of power, with coal's share falling to 60% by 2030.

The five-stage government plan in 2009 envisaged legislation for a regulatory framework in 2010 (passed in May 2011), investor, site, technology and construction arrangements over 2011-13, technical plans and site works 2014-15, construction of the first unit 2016-20 and successive units constructed by 2030.

In order to deliver the government's objectives, PGE, as Poland's largest power group by generating capacity, announced in January 2009 plans to build two nuclear power plants, each with a capacity

of 3000 MWe, one in the north, probably at Zarnowiec, and one in the east of the country.

The Ministry of Economy set out a new nuclear power programme in November 2010, and this was approved by the government in January 2011, and confirmed by the PGE board in February 2012. PGE estimated then that the cost would be €2500-3000/kW for a modern plant. It estimated the levelized cost of generating electricity from nuclear power plants at between €6.5 and €6.8 cents per kWh, which "justifies construction of plants under most scenarios."

In May 2011 parliament decisively passed – by 407 votes to 2 – legislation amending the country's Nuclear Energy Law to “provide for the establishment of a transparent and stable regulatory framework covering the entire investment process” by the National Atomic Energy Agency (Państwowa Agencja Atomistyki, PAA), which would oversee construction of the plants. It covers plant operation and the management of radioactive waste and used fuel.

In November 2018, the Ministry of Energy published its draft Energy Policy of Poland until 2040 (EPP2040) for public consultation, then estimating the cost of constructing nuclear at €4660/kWe. It was adopted by the country's cabinet in February 2021, and reaffirms plans to develop 6-9 GWe of nuclear energy. The first of six 1-1.5 GWe units was then planned to be in operation in 2033, with five successive units to follow every 2-3 years.

In June 2019, Poland signed a bilateral agreement on civil nuclear cooperation with the USA. This was followed in March 2021 by the ratification of an intergovernmental nuclear cooperation agreement that gives the USA 18 months to prepare a technology and financing offer for nuclear power plants. In June 2021 the US Trade & Development Agency provided a grant to PEJ to assist front-end engineering and design studies by Westinghouse and Bechtel with a view to building an AP1000 reactor as the country’s first nuclear power plant.

In April 2021 the state treasury took over PGE EJ1 for PLN 531 million ($140 million) and a new state-owned company, Polish Nuclear Power Plants (Polskie Elektrownie Jądrowe, PEJ), was set up to lead the investment. It would hold 51% of the construction company and project to about PLN 80 billion ($21 billion), with a partner for 49% being sought. This followed PGE reiterating in April 2020 that it would be unable to finance the plant from its own balance sheet. An account of earlier financing and commercial arrangements is given in the Appendix below. The government has

held talks with several potential partners for the investment, with the expectation that a foreign partner would hold a significant stake as well as supplying the reactor technology.

In October 2021 EDF offered to build up to six 1650 MWe EPR units. In April 2022 Korea Hydro & Nuclear Power (KHNP) submitted an equivalent offer to build six of its 1345 MWe APR1400 units, stating the first reactor could be in operation by 2033. KHNP has said it is willing to finance 20-30% of the project. In September 2022 Westinghouse and Bechtel submitted an offer to build six large-scale reactors in the country.

At the end of October 2022 prime minister Mateusz Morawiecki announced that Westinghouse had been chosen to build the country's first nuclear plant in the Pomerania province, with the exact location near the Baltic Sea to be confirmed. An additional agreement laying out the next steps – including site layout, licensing and permitting support, engineering services contracts and procurement and construction planning services – was signed between Westinghouse and PEJ in December 2022.

Also at the end of October 2022, energy company ZE PAK announced it had signed a letter of intent with PGE and Korea Hydro & Nuclear Power (KHNP) to build a nuclear plant based on APR1400 technology at the Pątnów site in central Poland.

In November 2022 KHNP commenced a site assessment at Pątnów. The following month, four eastern German states – Mecklenburg-Western Pomerania, Brandenburg and Saxony, which border Poland, as well as Berlin – formally expressed opposition to the Polish government's plans to develop the country's first nuclear power station.

In February 2023 PEJ concluded an agreement with the AGH University of Science and Technology in Warsaw for cooperation in the development of nuclear technologies and other technologies applicable in the nuclear industry, as well as activities supporting the development of nuclear energy in Poland.

In April 2023 ZE PAK and PGE announced the establishment of a joint venture known as PGE PAK Energia Jądrowa to manage the development of a nuclear plant in Pątnów. In October 2025 PGE announced that it would acquire the 50% owned by ZE PAK.

Earlier in July 2023 Poland's Ministry of Climate and Environment gave a decision-in-principle for PEJ to construct a nuclear power plant consisting of three Westinghouse AP1000 units in Pomerania

In January 2024 the government approved nearly $15 billion (approximately 30% of the planned cost) in finance for the country's first nuclear power plant. In September the government confirmed its intention to provide state guarantees covering 100% of debt taken by PEJ to finance the investment project; and a two-way contract for difference (CfD) providing revenue stability over the 60-year operating lifetime of the power plant – in addition to the equity injection announced in January. In December 2024 the European Commission launched an investigation into whether this planned support was compliant with EU rules on state aid. In December 2025 the Commission approved the state aid package, with Poland committing to reduce the CfD period from 60 to 40 years.

Earlier in March 2025 Polish President Andrzej Duda signed a bill designating that PEJ would receive the public financing of about $15 billion between 2025 and 2030. In December 2025, PEJ received PLN 4.6 billion (approximately $1.3 billion) in treasury bonds from the state.

Site selection and infrastructure

In 2013, an IAEA integrated nuclear infrastructure review (INIR) mission was undertaken. A follow-up to this in June 2016 concluded that all the recommendations and suggestions from 2013 regarding infrastructure had been implemented. These covered safety, waste, security and non-proliferation as well as coordination of operational and regulatory functions. An IAEA integrated regulatory review service (IRRS) mission was undertaken at the invitation of the government early in 2013 to scrutinize the regulatory structures set up in 2011-12. It also identified the need to upgrade grid infrastructure.

Site characterization work was undertaken at Zarnowiec, Choczewo and Lubiatowo-Kopalino in Pomerania. The Zarnowiec site, in the Krokowa and Gniewino administrative districts, is on a lake about 10 km from the Baltic coast, and is where construction of a nuclear plant started in 1980s. The Lubiatowo-Kopalino and Choczewo sites are within 2 km of each other on the coast in the Choczewo administrative district. The Choczewo site was rejected on social and environmental grounds.*

* A 2009 shortlist had Zarnowiec, Kopan and Lubiatowo/Klempicz as most likely. Sites in Nowe Miasto and Pilica (Mazovia province) and Belchatow were also being considered. Gaski in West Pomerania was considered from 2011 but dropped off the list

about 2013. In January 2019 Belchatow was mentioned as a possible site for the second plant.

Following a public tender process, in January 2013 PGE EJ1 awarded an $81.5 million contract to WorleyParsons to carry out site characterization, licensing and permitting for the country's first nuclear power plant. It expected this work to take until 2016. However, in December 2014 PGE EJ1 terminated the contract, citing slow progress, and took over the task itself. This delayed the project further. In April 2017 PGE EJ1 announced the start of “localization and environmental studies” to be carried out by ELBIS (a subsidiary of PGE Capital Group) at Zarnowiec and Lubiatowo-Kopalino.

Lubiatowo-Kopolino

The Lubiatowo-Kopolino site lies in Choczewo municipality in the Pomerania region on Poland’s Baltic coast. It is planned to host three 1250 MWe Westinghouse AP1000 units.

In December 2021 PEJ (formerly PGE EJ1 prior to its acquisition by the government) announced that Lubiatowo-Kopalino had been selected as the preferred location for hosting the country’s first plant.

In October 2023 the governor of the Pomeranian voivodeship issued a decision to determine the territorial scope of the Lubiatowo-Kopalino site. The decision allowed PEJ to carry out preparatory activities at the site. In January 2024 the Polish energy grid operator, PSE, announced that it had established the necessary criteria to link the proposed plant to the national grid.

Construction of the new power plant would necessitate a dock, 26 km of new roads, 30 km of new railway, upgrades to 66 km of railway, a new station, as well as a major grid connection.

PEJ signed an engineering services contract with Westinghouse and Bechtel in September 2023 to finalize the design of three AP1000 units at Lubiatowo-Kopalino. The agreement expired at the end of March 2025, but in April a further engineering development agreement was signed for the next nine months.The Pomeranian voivodeship issued a permit to PEJ to begin the first phase of site preparations including site clearance, levelling and landscaping in August 2025 and in September 2025 a further approval enabled geotechnical surveying and site development.

In December 2025 the European Commission approved a state aid package to support the construction and operation of the nuclear power plant. The approved support includes a two-way contract for difference providing revenue stability for 40 years with the Polish state to compensate PEJ when market prices fall below an agreed strike price.

In January 2026 Bechtel and Westinghouse announced a decision to use Arabelle steam turbines and generator sets for the three AP1000 units.

PEJ expects to be issued the construction permits that allow nuclear construction to begin in 2028. Two construction permits are needed, one from the National Atomic Energy Agency (Państwowa Agencja Atomistyki, PAA) as well as one from the Pomeranian voivodeship.

PEJ projects that the first unit could start up in 2036 with the other two following in 2037 and 2038.

Small modular reactors

Poland has a number of energy-intensive industrial companies – including Synthos, Ciech, KGHM and Orlen – working towards upgrading plants to include small reactors.

In August 2021, Synthos Green Energy began screening sites for small modular reactors (SMRs). Synthos Green Energy is a subsidiary of Synthos, a chemical manufacturing company headquartered in Poland. It has signed agreements related to SMR development with GE Hitachi Nuclear Energy and Ultra Safe Nuclear Corporation. It was collaborating with ZE PAK to assess the feasibility of replacing coal units at the Pątnów power plant with nuclear units, but this did not proceed.

In September 2021 NuScale started exploring with liquid fuels trading group Unimot and copper and silver producer KGHM Polska Miedź possibilities for its reactors to replace coal-fired power plants in Poland. In April 2023 KGHM submitted an application to the Ministry of Climate and Environment for the construction of a NuScale VOYGR modular nuclear power plant with a capacity of 462 MWe consisting of six 77 MWe modules. The application was approved in July 2023.

In January 2023 Polish renewable energy trader Respect Energy signed an agreement with EDF to cooperate on the development of nuclear power projects in Poland based on France's Nuward SMR technology. The technology is in the conceptual design phase, with the basic design phase anticipated to be completed by 2025 – so

the construction of a demonstration Nuward SMR could start in 2030.

In February 2023 Orlen's CEO, Daniel Obajtek, announced the company's amibition to construct 76 SMRs in 26 locations by 2038, with the first SMR to be built in 2028. Later that month, Polish state-owned group Industria signed a memorandum of intent with Rolls-Royce SMR as part of the 'Central Hydrogen Cluster' objective, which plans to produce 50,000 tonnes of low-carbon hydrogen each year.

In April 2023 Orlen Synthos Green Energy (OSGE) – a joint venture between Orlen and chemical producer Synthos Green Energy – announced seven possible sites for the deployment of GE Hitachi BWRX-300 units (Ostrołęka, Włocławek, Stawy Monowskie, Dąbrowa Górnicza, Nowa Huta, Stalowa Wola-Tarnobrzeg Special Economic Zone, and Warsaw). In May 2023 the country's regulator, the National Atomic Energy Agency (PAA), issued a positive general opinion which confirmed the validity of the BWRX-300 design assumptions and its compliance with Polish nuclear safety and radiological standards. In June 2023 the General Director of Environmental Protection started proceedings to issue the environmental decision for the construction of SMRs in Stawy Monowskie. The same proceedings were taken for the projects in Włocławek and Ostrołęka in August and early September. In December 2023 the government granted a decision-in-principle for the construction of up to 24 BWRX-300 reactors at these sites.

In January 2024 Orlen announced plans to build 600 MWe of SMR capacity by 2035. In February 2025 the General Director of Environmental Protection issued a scope of environmental reports for the proposed projects at Włocławek and Ostrołęka, allowing OSGE to begin environmental and siting research. The same was issued for Stawy Monowskie in February 2024.

In August 2025, Orlen and Synthos Green Energy announced that Włocławek would be the first reactor site to be developed.

High-temperature reactors

Deployment of high-temperature gas-cooled reactors (HTRs) for industrial heat production was included in the government's July 2016 draft strategy for development. The Ministry of Energy has estimated that using nuclear high temperature heat for industrial applications could reduce carbon dioxide emissions by 14-17

million tonnes per year in Poland, which has 13 large chemical plants that need 6500 MWt at 400-550°C.

A research project is being carried out at the National Centre for Nuclear Research (NCNR).

Public opinion

Public support for nuclear in Poland is strong. For more information see information page on Nuclear Energy and Public Opinion.

In December 2024 a poll from Polish research agency CBOS commissioned by Poland's ministry of climate found that more than 92% of respondents support the construction of nuclear power plants in the country. Support in the annual poll has risen steadily since it began in 2012.

In August 2022 ARC Market and Opinion's survey of 1037 Polish residents found that 64% of respondents supported the use of nuclear power, with only 13% opposed. The company suggested the 7% reduction in opposition to nuclear from the previous year's survey might be down to rising electricity prices and concerns over energy security.

A Polish Institute of International Affairs (PISM) survey in August 2014 (N=1000) showed public support for building the country's first nuclear power plant at 64%. Of these, 57% cited its potential for providing increased energy independence for the country as a reason for their support. Economic benefits were less frequently cited: 42% of pro-build respondents cited employment opportunities, while 26% and 24% respectively cited technological progress or the involvement of Polish companies in the project. Some 63% of those in favour of the nuclear option said they would support investment in nuclear capacity for Poland even if the country could meet its energy demand by buying low-priced power from its neighbours. The study found that the most prominent group among those supporting the construction of a nuclear plant was young, highly-educated individuals with higher incomes living in the largest cities. The study also found a clear regional pattern, with the highest support for nuclear coming from those in the east of the country. Previous polls conducted by Poland's independent Public Opinion Research Centre (CBOS) showed maximum support of 50%, in 2009.

Fuel cycle

The state-owned Radioactive Waste Disposal Enterprise (*Zakład Unieszkodliwiania Odpadów Promieniotwórczych*, ZUOP) has been operating for many years in connection with two research reactors. ZUOP operates the National Radioactive Waste Repository (*Krajowe Składowisko Odpadów Promieniotwórczych*, KSOP) in Różan. In August 2023 Poland's Ministry of Climate and Environment announced it is seeking municipalities interested in hosting a new repository for low- and intermediate-level short-lived waste.

In 1998 ZUOP identified possible sites for a deep geological repository, and was planning to establish an underground research laboratory to prepare for long-term placement of used fuel. The National Atomic Energy Agency (PAA) at the time nominated five potential locations: Lanieta, Klodawa, Damaslawek, Jarocin and Pogorzel.

Research & development

The National Centre for Nuclear Research (Narodowe Centrum Badań Jądrowych, NCBJ) has a 30 MWt multi-purpose research reactor – Maria – in operation. This started up in 1974 at Swierk, south of Warsaw. It was modified in the 1980s and resumed regular operation in 1993 as part of a European network of reactors for production of molybdenum-99. In 2012 the centre proposed making this, at least in its materials test role, a satellite of the 100 MWt Jules Horowitz Reactor (JHR) in France. JHR is under construction by the CEA at Cadarache in southern France. It is being built and will be operated in the framework of a 'consortium agreement' among several organizations, including Spain's Ciemat, Belgium's SCK, Czech Republic's NRI, Finland's VTT, Israel's IAEC, India's DAE, and Japan's JAEA. NCBJ aspires to join this team.

In July 2025, the National Atomic Energy Agency (PAA) granted an indefinite permit for Maria's operation. The permit specifies detailed operating conditions for the reactor. The operation of the reactor,

including compliance with the permit conditions, would be checked during regular inspections by the PAA, as well as during periodic safety assessments.

An earlier 10 MW research reactor, Ewa, started up in 1958 and was shut down in 1995.

Regulation, safety & non-proliferation

The Nuclear Energy Law was amended in 2011 to "provide for the establishment of a transparent and stable regulatory framework covering the entire investment process" by PAA. It reports to the Minister of Environment.

The PAA has signed cooperation agreements with the US Nuclear Regulatory Commission (NRC), the French *Autorité de sûreté nucléaire* (ASN), and the South Korean Nuclear Safety and Security Commission (NSSC).

In March 2010 a nuclear cooperation agreement related to nuclear safety and a legal and regulatory framework for a nuclear industry was signed with Japan. In July 2010 a nuclear cooperation agreement was signed with the USA, and in July 2017 one was signed with China's National Energy Administration.

Poland joined the Global Nuclear Energy Partnership (GNEP, now the International Framework for Nuclear Energy Cooperation, IFNEC) in September 2007 and the OECD Nuclear Energy Agency (NEA) in November 2010.

Non-proliferation

The application of safeguards in Poland under the NPT safeguards agreement INFCIRC/179, in force since October 1972, was suspended on 1 March 2007, on which date the Euratom agreement of April 1973 entered into force for Poland.

Notes & references

Sources

1. International Energy Agency, Energy Statistics Data Browser [Back]

General references

D.W.Kulczynski, Planning for a Nuclear Poland, Nuclear Engineering International, April 2014

Appendix

Earlier financing and commercial arrangements

PGE originally aimed to hold 51% of the projects through PGE Energia Jadrowa (PGE EJ) as part of a consortium with foreign strategic partners which can bring finance from their export credit agencies (ECAs). PGE would draw up agreements with potential strategic investors once the technology is chosen, since some contenders are closely linked to reactor vendors. PGE planned to finance 15% of the project from its own equity, with strategic partners taking another 15%, while 50% would be sought from ECAs and "perhaps 20%" being commercial debt. PGE EJ was set up in 2009 to develop nuclear power within the PGE Capital Group. PGE aims to have one main contractor, responsible for nuclear island, conventional island, civil engineering, balance of plant and site civil works, in line with IAEA guidelines.

In January 2010 PGE EJ1 was then set up as a limited liability company with 51% equity from PGE EJ and 49% PGE. It was responsible for the investment process, site selection, permitting and then building and operating the first nuclear power plant.

In July 2012 copper miner KGHM Polska Miedz (31.8% state-owned) and power utilities Tauron Polska Energia (30% state-owned) and Enea (51.5% state-owned) agreed to take minority stakes in the $10.3 to $11.3 billion project with PGE. A letter of intent was signed in September and in June 2013 the four companies signed an agreement to continue efforts towards "a draft purchase agreement of shares in the special purpose vehicle established for the construction and operation of an atomic power plant." The need for state financial support for the project was widely acknowledged, but the Treasury maintained that it was unaffordable. In September 2013 PGE confirmed that it would maintain 70% equity in PGE EJ1, with 10% each being held by

Enea, Tauron and KGHM, and all four parties initialled an agreement accordingly. This was confirmed in September 2014, with commitment to fund PLN 1 billion (about $310 million) towards the first phase of 3 GWe. In April 2015 there was a further announcement that the three companies had acquired the 30% equity in PGE EJ1, with PGE retaining 70%. In May 2018 it was reported that the three minor shareholders would withdraw from the PGE EJ1 consortium.

PGE expected to sign a commercial contract about the end of 2013 for the first 3000 MWe plant, so that first concrete would be poured in 2016. However, as it became apparent that the EPC contract could not be separated from the choice of a strategic partner and a financing model, PGE decided to set aside the traditional separate EPC contract package approach and look at wider options, including a more integrated contracting model and a broader scope for the contract. A revised programme, with schedule for the first unit to start up in 2024, was endorsed by the Council of Ministers in January 2014. Each unit was expected to cost PLN 50-60 billion (€12-14 billion).

In February 2014 four bidders submitted tender offers to PGE EJ1 to provide technical assistance as owner's engineer for the programme. These were AMEC Nuclear UK, Exelon Generation, a Mott MacDonald-AF Consult consortium and a URS Polska-Tractebel Engineering consortium. In July the company announced its selection of AMEC Nuclear UK, now AMEC Foster Wheeler. The owner's engineer would help select the EPC contractor, oversee project management and supply chain contract management, as well as regulatory aspects.

PGE was expected to make a final investment decision on the two units early in 2017, including site and technology, allowing construction start in 2020. The first unit was then expected to be operational in 2024, the second in 2029. However, in mid-2015 the transmission operator said that PGE projected 2029 for the first unit, with the second one scheduled for operation in 2035. PGE estimated then that conditional contracts would be signed in mid-2019, so the decision on technology and vendor was expected in 2018-19.

Early in 2015 PGE told the Ministry of Economy that government-guaranteed long-term contracts for selling power were the best way to set up the project to enable financing. PGE said:

"Having described and justified a catalogue of potential support mechanisms, it had singled out contracts for difference (CfD) as the

mechanism that should be dedicated to nuclear energy... It is assumed that this type of mechanism should apply market tools in a manner similar to the contracts for difference mechanism used in the United Kingdom... PGE strongly believes that today's energy-only market does not support such capital intensive investments as new nuclear build, and CfD is the best way to minimize market risk for the investment. The CfD both provides the needed degree of certainty to the investor with regard to future revenues over a long timeframe, as well as protects the rate payers from potential overcompensation. The fact that this mechanism has already been accepted by the European Commission (EC) with regard to the British project also supports our view that it is the most appropriate instrument to use in our case." However, in June 2016 the new government rejected the use of CfDs as being too costly, and said that it favoured maintaining high dependence on coal.

In July 2017 a government delegation visited China General Nuclear Power Corporation (CGN) in Shenzhen, and CGN said it was very interested in "becoming a long-term strategic partner of Poland and helping with the localization of nuclear technology."

Earlier in November 2009 France and Poland signed a joint declaration on energy, environment and climate that called for France to assist Poland in the construction of nuclear power plants. PGE then signed an agreement to work with EdF to investigate using EPR technology for Poland, and Areva said that it would bid in conjunction with EdF. A similar non-exclusive agreement was signed with GE Hitachi early in 2010, regarding ABWR and ESBWR technology. Westinghouse has signed a similar agreement for its AP1000. Russian technology was not under consideration. PGE EJ1 said in November 2015 that five companies had expressed interest in bidding: GE Hitachi, Kepco, SNC-Lavalin, Westinghouse and EdF/Areva.

In October 2021 EDF submitted an offer to the Polish government to build four to six EPR reactors in the country.

Lithuanian liaison

In July 2006 Lithuania invited Poland to join with Estonia and Latvia in building a new large reactor in Lithuania, to replace the Ignalina units being shut down at EU insistence. Polish participation would justify a larger and more economical unit. In February 2007 the three Baltic states and Poland agreed to build a new nuclear plant there, initially of 3200 MWe. Lithuania as host would have 34% of the project and Poland, Latvia and Estonia 22% each. At least one

unit of the project was expected to be operating by 2015. Total cost would be some €6 billion. E.ON earlier expressed interest in investing in such a unit. Poland said that unless it had access to at least 1000 MWe of the project, later increased to 1200 MWe, it would not be worth building the transmission lines to Poland.

In July 2008 the Lithuanian government with energy companies from Latvia, Estonia and Poland (Latvenergo, Eesti Energia and Polska Grupa Energetyczna) established the Visaginas project development company Visagino Atomine Elektine (VAE) for the new 3200-3400 MWe nuclear power plant. Lithuania held 51% of this, and the others 16% each, but the JV was to be reconstituted later as a project implementation company with different share splits related to long-term equity. Though located close to the Soviet-era Ignalina plant, the new one will be called Visaginas after the nearby town of that name. Lithuania wanted at least 34% of the new plant (1090-1160 MWe), Poland wanted 1000 MWe, while Latvia and Estonia wanted 400-600 MWe each. However, only one reactor was then envisaged, at a cost of about €4 billion, and completion date slipped well beyond 2020. In December 2011, PGE withdrew from the project, saying that VAE's conditions were unacceptable to PGE, and that it wanted to focus on its own plans. PGE also said it would not buy any power from Russia's (now stalled or aborted) Baltic plant in Kaliningrad. (Further details in the information paper on Lithuania).